

BM
3/31/22

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

PRECIPITATION SAMPLES

DATE COLLECTED: 21 MAR 2022 Initials TW

Conversion Factors

10 in. Bulk Rain Funnel: NET divided by 51.1 = catch

N/A if Overfilled

12 in. Bulk Snow Bucket: NET divided by 69.3 = catch

SITE	TIME (EST)	COLLECTOR VOLUME (ml)	CATCH (mm)	ETI CATCH (mm)	Coll. EFF. (% of ETI)
RG-22	<u>0730</u>	<u>820</u>	<u>11.8</u>		
RG-1	<u>0940</u>	<u>830</u>	<u>12.0</u>		
RG-4	<u>1000</u>	<u>896</u>	<u>12.9</u>		
RG-23	<u>0835</u>	<u>936</u>	<u>13.5</u>		

Volumes Collected				
	RG-22	RG-1	RG-4	RG-23
Gross g	<u>1781</u>	<u>1790</u>	<u>1856</u>	<u>1897</u>
Tare g	<u>961</u>	<u>960</u>	<u>960</u>	<u>961</u>
Net g	<u>820</u>	<u>830</u>	<u>896</u>	<u>936</u>

SITE	CONTAMINATION (0 = none. 1 = low -----> 5 = high)						dBASE FIELD code?	SAMPLE REJECTED @LEVEL =	Database CODES checked ?
	Fine PM	Coarse PM	Bird FECES	Insects	Color	GB Ring			
RG-22	<u>2</u>	<u>2</u> Bark	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>			
RG-1	<u>2+</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>			
RG-4	<u>2</u> Hemlock needles	<u>2</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>			
RG-23	<u>2+</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>			

Precipitation amount
.65 inches / 16.4 mm
ETI collector

ADDITIONAL COMMENTS: Precip was mostly rain, with some snow. Liquid when collected.
SG-22 = 3" Bare spots SG-1 = 0", patchy SG-4 = 10" SG-23 = 16"
FS data = 2-38cmD, 0-18cm SWE

Applicable dBASE FIELD codes:
(See SOP Manual)

970 = Obvious Contamination: Sample REJECTED

969 = Possible Contamination: Check Return Analyses

955 = QNS: Quantity Not Sufficient for analyses (< 100 mL NET catch)

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

Streamwater SamplesDate Collected: 21 MAR 2022Initials: TW

SITE	TIME (EST)	TEMP (°C)	GAGE HT.	HYDROGR L/R/F/PK	COLOR (0 = none · 1 = low → 5 = high)	SEDIMENT	REMARKS	dBASE FIELD code?
WS-1	<u>0940</u>	<u>2.5°</u>	<u>.470</u>	<u>F</u>	<u>0</u>	<u>0</u>	<u>5% sc</u>	<u>opw</u>
WS-2	<u>0950</u>	<u>3.4°</u>	<u>.456</u>	<u>F</u>	<u>0</u>	<u>1</u>	<u>0-25% sc</u>	
WS-3	<u>1010</u>	<u>3.6°</u>	<u>.655</u>	<u>F</u>	<u>1-</u>	<u>0</u>	<u>25-50% sc</u>	
WS-4	<u>0930</u>	<u>2.2°</u>	<u>.634</u>	<u>F</u>	<u>0</u>	<u>1</u>	<u>0-25% sc</u>	
WS-5	<u>0910</u>	<u>2.0°</u>	<u>.633</u>	<u>F</u>	<u>0</u>	<u>0</u>	<u>50% sc</u>	
WS-6 (Dup)	<u>0920</u>	<u>2.1°</u>	<u>.513</u>	<u>F</u>	<u>1-</u>	<u>0</u>	<u>25-50% sc</u>	<u>opw</u>
WS-7	<u>0850</u>	<u>1.0°</u>	<u>.826</u>	<u>F</u>	<u>0</u>	<u>1</u>	<u>Blown open</u>	
WS-8	<u>0840</u>	<u>0.8°</u>	<u>.705</u>	<u>F</u>	<u>1-</u>	<u>0</u>	<u>mostly opw steep bank ice</u>	<u>99% sc</u>
WS-9	<u>0825</u>	<u>0.5°</u>	<u>.704</u>	<u>F</u>	<u>2-</u>	<u>0</u>	<u>weir fzn/foam, stream mostly sc/ice</u>	
Hubbard BK	<u>1035</u>	<u>3.3°</u>		<u>F</u>	<u>1</u>	<u>0</u>	<u>Blown out! Bank ice intact</u>	
Mlake Outlet	<u>1100</u>	<u>4.6°</u>		<u>F</u>	<u>0</u>	<u>0</u>	<u>opw stuff, 0-1 B</u>	

ADDITIONAL COMMENTS: NF = 99% sc, South side is going fast (0-50%)
Falling hydrographs from weekend rain, melt.

Applicable dBASE FIELD codes 911 = High Flow Event in Progress, 912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)

955 = QNS: Quantity Not Sufficient for analyses

master sample notes 2017

HUBBARD BROOK ECOSYSTEM STUDY

METROHM / 3 STAR LAB ANALYSES

DATE RUN: 24 MAR 2022

RUN By: TW

Meter CALIBRATION: See Calibration form for Standard Procedure

Conductivity Correction Factor = 1.12 x

at: 19.3 °C Lab Temp

Samples:	DATE COLLECTED	Metrohm pH	3 Star pH	SpCond @ 25°C	Remarks Codes
Precipitation					
STATION	<u>21 MAR 2022</u>	<u>6.04</u>	<u>6.01</u>	<u>6.4</u>	
RG-1		<u>6.10</u>	<u>6.08</u>	<u>6.7</u>	
RG-4		<u>5.78</u>	<u>5.81</u>	<u>6.3</u>	
RG-23	<u>21 MAR 2022</u>	<u>6.03</u>	<u>5.97</u>	<u>5.5</u>	

	Metrohm pH	Metrohm ANC	ANC end pt. pH	3 Star pH	SpCond @ 25°C	Remarks Codes
Streamwater						
WS-1	<u>21 MAR 2022</u>	<u>5.44</u>	<u>+5.1</u>	<u>5.17</u>	<u>5.44</u>	<u>15.2</u>
WS-2		<u>5.50</u>	<u>+5.2</u>	<u>5.14</u>	<u>5.49</u>	<u>8.2</u>
WS-3		<u>5.51</u>	<u>+4.4</u>	<u>5.10</u>	<u>5.50</u>	<u>10.0</u>
WS-4		<u>5.84</u>	<u>+12.2</u>	<u>5.33</u>	<u>5.88</u>	<u>8.6</u>
WS-5		<u>5.40</u>	<u>+4.1</u>	<u>5.16</u>	<u>5.37</u>	<u>9.5</u>
WS-6		<u>5.32</u>	<u>+0.2</u>	<u>4.68</u>	<u>5.30</u>	<u>10.5</u>
WS-7		<u>5.54</u>	<u>+7.2</u>	<u>5.27</u>	<u>5.54</u>	<u>9.4</u>
WS-8		<u>5.29</u>	<u>+7.0</u>	<u>5.23</u>	<u>5.31</u>	<u>10.1</u>
WS-9		<u>4.61</u>	<u>-29.4</u>	<u>4.30</u>	<u>4.60</u>	<u>15.6</u>
Hubbard BK		<u>5.96</u>	<u>+16.9</u>	<u>5.33</u>	<u>5.95</u>	<u>9.4</u>
Mlake Outlet	<u>21 MAR 2022</u>	<u>6.36</u>	<u>+67.2</u>	<u>5.05</u>	<u>6.32</u>	<u>29.7</u>

ADDITIONAL COMMENTS: Ran WS 2x
New computer set-up to run titration system.

*) Precip values may be higher than if they were run earlier.
Sat in buckets for 3 days Snowmelt chemistry

Applicable codes: 911= High Flow Event in Progress;

955= QNS: Quantity Not Sufficient for analyses(<100 mL NET catch)

912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)

metrohm/3 star lab analyses 2017 v4

HBEF Stream and M.Lake Routine Sample DIC Datadate: **22MAR2022**

Samples collected 03/21/2022

Aqueous STANDARDS

type	DIC	peak ht	adjusted	ATT -X
	set =	mm	mm	
DIW blank	0	9	0	16
	0	10	0	16
	0	10	0	16
100 µM	100	30	21	16
	100	29	19	16
	100	30	20	16
200 µM	200	51	42	16
	200	51	41	16
	200	50	40	16
400 µM	400	100	91	16
	400	98	88	16
	400	98	88	16

Statistics

Values from matrix at M5 -->

slope = 4 463

slope ± 0 075

Y-int = 7.6

Y-int ± 3.7

R-sq = 0.9972

Laboratory Conditions

Lab Temp =

19.4 °C

Barom Pressure =

769 mm Hg

Analyst's Initials.

TW

48%RH

NaCO3 Standards

0 1M Stock Dated

3/10/22

0 01M Working Solution Dated.

3/10/22

Samples

	inj 1	inj 2	inj 3	ATT -X	Avg Peak	±SD Peak	Samples	DIC	Calc	NOTES
	1	2	3					µM/L	err±	120 mA current
ML-70	46	47	46	16	46.3	0.6	ML-70	207	3	
	0	0	0	16	0.0	0.0	0	0	0	
	0	0	0	16	0.0	0.0	0	0	0	
	0	0	0	16	0.0	0.0	0	0	0	Falling
	0	0	0	16	0.0	0.0	0	0	0	H-graphs
										Rain
W-1	18	18	22	16	19.3	2.3	W-1	86	10	
W-2	13	13	16	16	14.0	1.7	W-2	62	8	
W-3	11	12	13	16	12.0	1.0	W-3	54	4	
W-4	12	12	13	16	12.3	0.6	W-4	55	3	
W-5	10	11	11	16	10.7	0.6	W-5	48	3	
W-6	10	11	11	16	10.7	0.6	W-6	48	3	
W-7	13	13	13	16	13.0	0.0	W-7	58	0	
W-8	10	10	11	16	10.3	0.6	W-8	46	3	
W-9	11	11	11	16	11.0	0.0	W-9	49	0	
HBk	11	11	12	16	11.3	0.6	HBk	51	3	
	0	0	0	16	0.0	8.0	0			
							0			
Lab Air	15	15	15	16	15.0	0.0	Lab Air	67	0	
Outside Air	13	12	12	16	12.3	0.6	Outside Air	55	3	
Gas 1000	25	25	25	16	25.0	0.0	Gas 1000	112	0	std=83.3
Gas 3000	67	71	72	16	70.0	2.6	Gas 3000	312	12	std=246.3